Math 110 — Week 2

Linear independence, span, bases, dimension (Notes Ch. 3–4, LADR 2A–2C, HW2)

§1 Facts used from Week 1

(W1) A subset $U \subseteq V$ is a subspace if $0 \in U$ and $\lambda v + w \in U$ for all $v, w \in U$, $\lambda \in \mathbb{F}$. Intersections and sums $U + W = \{u + w\}$ of subspaces are subspaces.

(W2) The sum $U_1 + \cdots + U_m$ is direct if $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for each k. Equivalent: the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is with all $u_k = 0$ (Prop. 7). For two subspaces: $U \cap W = \{0\}$.

(W3) If $av = 0$ then $a = 0$ or $v = 0$.

§2 This week

§2.1 Linear independence, span, basis (Ch. 3)

The question. When does a finite list of vectors describe a whole vector space? Two things are needed: every vector can be built from the list, and in only one way. In $\mathbb{R}^2$: $(1,0), (0,1)$ does both. $(1,0), (0,1), (1,1)$ builds everything but not uniquely:

$$(1,1) = 1 \cdot (1,0) + 1 \cdot (0,1) = 1 \cdot (1,1).$$

$(1,0)$ alone builds uniquely but misses $(0,1)$.

Definition (Def. 7, 8 / LADR 2.4, 2.15, 2.26, 2.29)**.** Let $v_1, \ldots, v_m \in V$. A **linear combination** is a sum $\lambda_1 v_1 + \cdots + \lambda_m v_m$ with $\lambda_k \in \mathbb{F}$.

- $\operatorname{span}(v_1, \ldots, v_m)$ is the set of all linear combinations.
- The list **spans** V if $\operatorname{span}(v_1, \ldots, v_m) = V$.
- The list is **linearly independent** if $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ forces every $\lambda_k = 0$. Otherwise it is **linearly dependent**.
- The list is a **basis** of V if it is linearly independent and spans V.
- V is **finite-dimensional** if some finite list spans it.
- The **length** of a list is its number of entries. The empty list has length 0, is linearly independent, and spans $\{0\}$.
- A list containing 0 is dependent: $1 \cdot 0 = 0$ with coefficient $1 \neq 0$.
- $\operatorname{span}(v_1, \ldots, v_m) = \mathbb{F}v_1 + \cdots + \mathbb{F}v_m$, where $\mathbb{F}v = \{\lambda v\}$. It is the smallest subspace containing the v_k (W1).
- Ex. 10: $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$. The span of $(1,1,1,\ldots)$ and $(0,1,2,3,\ldots)$ in $\mathbb{R}^\infty$ is the set of arithmetic sequences $(a_0, a_0 + d, a_0 + 2d, \ldots)$. The only basis of $\{0\}$ is the empty list.

Proposition (Basis = unique representation, LADR 2.28)

$v_1, \ldots, v_m$ is a basis of $V \iff$ every $v \in V$ can be written as $\lambda_1 v_1 + \cdots + \lambda_m v_m$ in exactly one way.

The idea

The statement has two halves, and each half of the definition of a basis takes care of one of them.

- "Every v can be written as $\lambda_1 v_1 + \cdots + \lambda_m v_m$" is, word for word, the definition of "the list spans V".
- "In only one way" is what independence gives. Suppose v could be written with coefficients λ_k and also with coefficients μ_k. Subtract one expression from the other: the left sides give $v - v = 0$, the right sides give $\sum(\lambda_k - \mu_k)v_k$. So we have a combination equal to 0, and independence says its coefficients $\lambda_k - \mu_k$ are all 0.

Example: in $\mathbb{R}^2$ with the basis $(1,1), (1,-1)$, the vector $(3,1)$ is $2(1,1) + 1(1,-1)$, and no other pair of coefficients works.

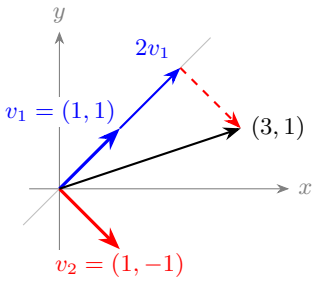
Proof. **($\Rightarrow$) Assume $v_1, \ldots, v_m$ is a basis.** Let $v \in V$.

1. *v can be written as a combination.* The list spans V, which means exactly that every vector of V is a linear combination of $v_1, \ldots, v_m$.
2. *Only one way.* Suppose $v = \lambda_1 v_1 + \cdots + \lambda_m v_m$ and also $v = \mu_1 v_1 + \cdots + \mu_m v_m$. Subtracting, $0 = (\lambda_1 - \mu_1)v_1 + \cdots + (\lambda_m - \mu_m)v_m$.
3. The list is independent, so every coefficient $\lambda_k - \mu_k$ is 0. So $\lambda_k = \mu_k$ for all k: the two ways are the same.

($\Leftarrow$) Assume every $v \in V$ can be written as a combination in exactly one way.

1. *The list spans V.* Every v can be written as a combination; that is the definition of spanning.
2. *The list is independent.* Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$. We also have $0v_1 + \cdots + 0v_m = 0$. These are two ways of writing the vector 0. By assumption there is only one way, so $\lambda_k = 0$ for all k.

□



Go 2 steps along v_1, then 1 step along v_2. Spanning says every point can be reached this way; independence says there is only one way.

- The unique coefficients $\lambda_1, \ldots, \lambda_m$ are called the **coordinates** of v in the basis.
- The notes say the same thing with the map $\Phi\colon \mathbb{F}^m \to V$, $\Phi(\lambda_1, \ldots, \lambda_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m$. The list spans V exactly when Φ is surjective (every v is hit). It is independent exactly when Φ is injective (no v is hit twice). It is a basis exactly when Φ is bijective.
- Connection with direct sums: if all $v_k \neq 0$, the list $v_1, \ldots, v_m$ is independent exactly when the sum of the lines $\mathbb{F}v_1 + \cdots + \mathbb{F}v_m$ is direct. Reason: by (W2) the sum is direct when $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ forces every *term* $\lambda_k v_k$ to be 0, and by (W3) $\lambda_k v_k = 0$ with $v_k \neq 0$ means $\lambda_k = 0$.

Proposition (Prop. 8 / LADR 2.19: linear dependence lemma)

Let $v_1, \ldots, v_m$ be linearly dependent. Then

(i) there is an index j with $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$;

(ii) for any such j, removing v_j does not change the span.

The idea

Dependence gives an equation $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with some nonzero coefficient. Any vector with a nonzero coefficient can be solved for in terms of the others; that vector is redundant.

Why choose the *last* nonzero coefficient? Because then the other vectors in the equation all come *earlier* in the list. That is a stronger statement than "v_j is a combination of the others", and later proofs need it: when a list starts with vectors we want to keep, the redundant vector can be found after them.

Example: $(1,0), (0,1), (1,1), (2,0)$. One relation is $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) + 0 \cdot (2,0) = 0$. The last nonzero coefficient belongs to $(1,1)$, and $(1,1) = (1,0) + (0,1)$ is in the span of the vectors before it.

Proof. 1. Take $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with not all $\lambda_k = 0$. Let j be the largest index with $\lambda_j \neq 0$. The relation is $\lambda_1 v_1 + \cdots + \lambda_j v_j = 0$.

2. Divide by λ_j:

$$v_j = -\frac{\lambda_1}{\lambda_j} v_1 - \cdots - \frac{\lambda_{j-1}}{\lambda_j} v_{j-1} \in \operatorname{span}(v_1, \ldots, v_{j-1}).$$

(If $j = 1$ this says $v_1 = 0$, and span of the empty list is $\{0\}$.)

3. For (ii): write $v_j = c_1 v_1 + \cdots + c_{j-1} v_{j-1}$. In any combination $\lambda_1 v_1 + \cdots + \lambda_m v_m$, replace v_j by this expression. The result uses only the other vectors. So every vector in $\operatorname{span}(v_1, \ldots, v_m)$ is also in the span of the list without v_j. Conversely, a combination of the shorter list is a combination of the full list (give v_j the coefficient 0). So the two spans are equal.

□

Proposition (Prop. 9 / LADR 2.30: a spanning list contains a basis)

If $f_1, \ldots, f_n$ spans V, some of the f_k can be deleted so that what remains is a basis of V.

Proof. 1. If the list is independent, it is a basis.

2. If not, Prop. 8 gives a vector whose removal does not change the span. Remove it. The list still spans V and is one shorter.

3. Repeat. The list is finite, so this stops, and it stops only when the list is independent.

□

What has to be explained. Why can $\mathbb{R}^2$ not contain three independent vectors? There is no formula to compute; we need a mechanism that compares two lists.

Theorem (Prop. 10 / LADR 2.22: independent lists are not longer than spanning lists)
If $e_1, \dots, e_m$ is linearly independent in V and $f_1, \dots, f_n$ spans V, then $m \le n$.

The idea

The mechanism: one in, one out. Start with the spanning list $f_1, \dots, f_n$ and put e_1 into it. Since the f's span, e_1 is a combination of them, so the longer list $e_1, f_1, \dots, f_n$ is dependent, and one vector can be removed without changing the span. We want to remove an f, not e_1. That is possible: $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$ with some $\lambda_k \neq 0$ (because $e_1 \neq 0$), and that f_k can be solved for. Now we have a spanning list with e_1 and $n-1$ of the f's.

Repeat with e_2, e_3, ... Each time, the new e_k is a combination of the current list. Could that combination use only the e's already inserted? No: that would make $e_1, \dots, e_k$ dependent. So it uses some remaining f with a nonzero coefficient, and that f can be exchanged for e_k.

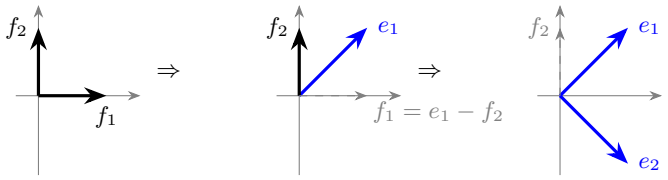
Why this gives $m \le n$. Each e that goes in uses up one f. If there were more e's than f's, the f's would run out: after n steps the list $e_1, \dots, e_n$ spans V, so e_{n+1} is a combination of $e_1, \dots, e_n$, contradicting independence.

Example in $\mathbb{R}^2$ with $e_1 = (1,1)$, $e_2 = (1,-1)$ and $f_1 = (1,0)$, $f_2 = (0,1)$: $e_1 = f_1 + f_2$, so exchange f_1: the list e_1, f_2 spans. Then $e_2 = e_1 - 2f_2$ uses f_2, so exchange it: e_1, e_2 spans. A third independent vector is impossible.

Proof. If $m = 0$ there is nothing to prove. Claim $P(k)$, for $1 \le k \le \min(m,n)$: after relabelling the f's, the list $L_k = (e_1, \dots, e_k, f_{k+1}, \dots, f_n)$ spans V.

1. $P(1)$: write $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$. Since $e_1 \neq 0$, some λ is nonzero; relabel so that $\lambda_1 \neq 0$. Then $f_1 = \frac{1}{\lambda_1}(e_1 - \lambda_2 f_2 - \cdots - \lambda_n f_n)$ is in $\operatorname{span}(e_1, f_2, \dots, f_n)$. So this span contains $f_1, f_2, \dots, f_n$. It is a subspace, so it contains every combination of the f's, and those are all of V.
2. $P(k-1) \Rightarrow P(k)$: since L_{k-1} spans, $e_k = \lambda_1 e_1 + \cdots + \lambda_{k-1} e_{k-1} + \mu_k f_k + \cdots + \mu_n f_n$.
3. Some μ is nonzero: otherwise $e_k \in \operatorname{span}(e_1, \dots, e_{k-1})$, contradicting independence of the e's. Relabel so that $\mu_k \neq 0$.
4. Solve for f_k in terms of $e_1, \dots, e_k, f_{k+1}, \dots, f_n$. So $\operatorname{span}(L_k)$ contains every vector of L_{k-1}, and L_k spans V.
5. Suppose $m > n$. Then $P(n)$ says $e_1, \dots, e_n$ spans V, so $e_{n+1} \in \operatorname{span}(e_1, \dots, e_n)$, contradicting independence. Hence $m \le n$.

The proof also shows: after all the e's are inserted, $e_1, \dots, e_m, f_{m+1}, \dots, f_n$ still spans V. □



Each step puts one e in and takes one f out, and the list still spans $\mathbb{R}^2$: the removed f (dashed) is a combination of the vectors kept, $f_1 = e_1 - f_2$ and then $f_2 = \frac{1}{2}(e_1 - e_2)$.

Proposition (Prop. 11 / LADR 2.32: an independent list extends to a basis)
If V is finite-dimensional and $e_1, \dots, e_m$ is linearly independent, there is a basis of V containing $e_1, \dots, e_m$. If $f_1, \dots, f_n$ spans V, the added vectors can be taken from the f_j.

The idea

Put the two lists together: $e_1, \dots, e_m, f_1, \dots, f_n$ spans V. Now remove redundant vectors as in Prop. 9. The only worry is that an e gets removed. The dependence lemma removes a vector that is a combination of the vectors *before* it, and the e's come first and are independent, so no e is a combination of the vectors before it. Only f's are removed.

Proof. 1. The list $e_1, \dots, e_m, f_1, \dots, f_n$ spans V.

2. If it is dependent, Prop. 8 gives a vector in the span of the earlier ones. It is not some e_i, because $e_i \notin \operatorname{span}(e_1, \dots, e_{i-1})$ by independence. So it is an f_j; remove it. The span is unchanged.
3. Repeat until the list is independent. It still spans V and still contains every e_i.

□

- Ex. 11: no three vectors in $\mathbb{F}^2$ are independent, since e_1, e_2 spans. This is why three lines in $\mathbb{R}^2$ never form a direct sum.
- **Every subspace has a complement (LADR 2.33).** If U is a subspace of finite-dimensional V, there is a subspace W with $V = U \oplus W$: extend a basis $u_1, \dots, u_k$ of U to a basis $u_1, \dots, u_k, w_1, \dots, w_l$ of V and let $W = \operatorname{span}(w_1, \dots, w_l)$. Then $U + W = V$, because every vector is a combination of the whole basis, and such a combination is (a combination of the u's) + (a combination of the w's). And $U \cap W = \{0\}$: if x is in both, then $x = \sum a_i u_i$ and $x = \sum b_j w_j$, so $\sum a_i u_i - \sum b_j w_j = 0$. The whole basis is independent, so all a_i and b_j are 0, and $x = 0$. (That U has a basis is Prop. 14 below.)

Infinite-dimensional spaces (Def. 9, optional in the notes). For an infinite set $M \subseteq V$: M is linearly independent if every finite subset is; M spans if every vector is a *finite* combination of elements of M; a basis is both.

- Ex. 12: $\{1, x, x^2, \dots\}$ is a basis of $\mathcal{P}(\mathbb{R})$.
- Ex. 13: in $\mathbb{F}^\infty$, $\{e_1, e_2, \dots\}$ is independent but not a basis: $(1, 1, 1, \dots)$ is not a finite combination (HW2 P2).

coordinate	1	2	3	4	5	$\cdots$
e_1	1	0	0	0	0	$\cdots$
e_3	0	0	1	0	0	$\cdots$
$2e_1 - e_3$	2	0	-1	0	0	$\cdots$
$(1, 1, 1, \dots)$	1	1	1	1	1	$\cdots$

The combination $2e_1 - e_3$ is zero after coordinate 3. Every finite combination of the e_j is zero after its largest index j, so none equals the all-ones sequence.

- Every vector space has a basis, but the proof for infinite-dimensional spaces needs Zorn's lemma.

§2.2 Dimension (Ch. 4)

Proposition (Prop. 12 / LADR 2.34, Def. 10)
Any two bases of a finite-dimensional V have the same length. This length is the **dimension** $\dim V$.

Proof. Let B_1 and B_2 be two bases. A basis is both independent and spanning. Use Prop. 10 with B_1 as the independent list and B_2 as the spanning list: $\operatorname{length} B_1 \le \operatorname{length} B_2$. Use it again with B_2 as the independent list and B_1 as the spanning list: $\operatorname{length} B_2 \le \operatorname{length} B_1$. So the lengths are equal. □

- Ex. 14–16: $\dim \mathbb{F}^n = n$. $\dim \mathcal{P}_r(\mathbb{F}) = r + 1$ (basis $1, x, \dots, x^r$). $\dim\{0\} = 0$.
- Ex. 17: the dimension depends on the field. $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, but $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$, with real basis $(1, 0), (i, 0), (0, 1), (0, i)$.

Proposition (Prop. 13 / LADR 2.38, 2.42: the right number of vectors)

Let $d = \dim V$. (i) Any linearly independent list of d vectors is a basis. (ii) Any spanning list of d vectors is a basis.

The idea

A basis has to pass two tests, independence and spanning. This proposition says that when the list already has exactly $\dim V$ vectors, passing one test is enough; the other follows. On an exam this halves the work.

Why it is true. We know two procedures: an independent list can be made into a basis by adding vectors (Prop. 11), and a spanning list can be made into a basis by deleting vectors (Prop. 9). The result of either procedure is a basis, so it has exactly d vectors. If the list we started with already had d vectors, the procedure cannot have added or deleted anything. So the list we started with was already a basis.

It is the same counting principle as in Week 1: a map between two finite sets of the same size is injective exactly when it is surjective.

Proof. 1. (i) Let $v_1, \dots, v_d$ be independent. By Prop. 11 we can add vectors to it (possibly none) to get a basis of V.

2. That basis has exactly d vectors, because every basis of V does (Prop. 12). The list $v_1, \dots, v_d$ already has d vectors, so no vectors were added. So $v_1, \dots, v_d$ is itself the basis.

3. (ii) Let $v_1, \dots, v_d$ span V. By Prop. 9 we can delete vectors from it (possibly none) to get a basis of V.

4. That basis has exactly d vectors, so no vectors were deleted. So $v_1, \dots, v_d$ is itself the basis.

□

Proposition (Prop. 14 / LADR 2.25, 2.37, 2.39: subspaces)

If U is a subspace of finite-dimensional V, then U is finite-dimensional, $\dim U \leq \dim V$, and $\dim U = \dim V \iff U = V$.

Proof. 1. *Independent lists in U cannot be long.* A list of vectors of U that is independent is also an independent list in V, so its length is at most $\dim V$ (Prop. 10). So among all independent lists in U there is one of largest length. Call it $u_1, \dots, u_k$; then $k \leq \dim V$.

2. *This list spans U.* Suppose not: some $u \in U$ is not in $\operatorname{span}(u_1, \dots, u_k)$. We show that then $u_1, \dots, u_k, u$ is independent, which contradicts the choice of k as the largest length.

3. Take a relation $\lambda_1 u_1 + \cdots + \lambda_k u_k + \mu u = 0$. If $\mu \neq 0$, solve for u: $u = -\frac{1}{\mu}(\lambda_1 u_1 + \cdots + \lambda_k u_k)$, so u would be in the span. So $\mu = 0$. The relation is then $\lambda_1 u_1 + \cdots + \lambda_k u_k = 0$, and all $\lambda_i = 0$ because $u_1, \dots, u_k$ is independent.

4. So $u_1, \dots, u_k$ is independent and spans U: it is a basis of U. Hence U is finite-dimensional and $\dim U = k \leq \dim V$.

5. *Equality.* If $\dim U = \dim V$, then $u_1, \dots, u_k$ is an independent list of $\dim V$ vectors in V, so it is a basis of V (Prop. 13). Then every vector of V is a combination of $u_1, \dots, u_k$, and those combinations are in U. So $V \subseteq U$, i.e. $U = V$. Conversely, if $U = V$ the dimensions are equal.

□

- Ex. 18: $1, x - a, (x - a)^2, \dots, (x - a)^n$ is a basis of $\mathcal{P}_n(\mathbb{R})$. Independent: the degrees are different, so in a relation the highest-degree term with nonzero coefficient cannot cancel. And there are $n + 1 = \dim \mathcal{P}_n(\mathbb{R})$ of them (Prop. 13). The coordinates are the Taylor coefficients: $p(x) = \sum_k \frac{p^{(k)}(a)}{k!}(x - a)^k$.
- Finite dimension is needed: $\mathcal{P}(\mathbb{R})$ has the proper subspace $\{p : p(0) = 0\}$, and both are infinite-dimensional.

Theorem (Prop. 15, 16 / LADR 2.43: dimension of a sum)

For subspaces U, W of a finite-dimensional space:

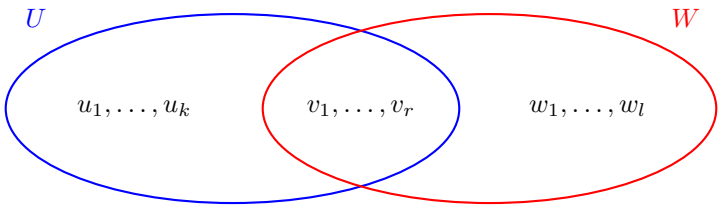
$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

More precisely, if $v_1, \ldots, v_r$ is a basis of $U \cap W$, it extends to a basis $v_1, \ldots, v_r, u_1, \ldots, u_k$ of U and to a basis $v_1, \ldots, v_r, w_1, \ldots, w_l$ of W, and then $v_1, \ldots, v_r, u_1, \ldots, u_k, w_1, \ldots, w_l$ is a basis of $U+W$.

The idea

For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$: adding the two sizes counts the common part twice. The same thing happens with bases, provided the bases of U and W are chosen so that their common part is a basis of $U \cap W$. That is why the proof *starts* from a basis of $U \cap W$ and extends it in two directions.

Putting the three groups together clearly spans $U+W$. The step that needs an argument is independence, and the question to ask is: in a relation, could the u-part and the w-part cancel each other? Move the w-part to the other side. Then one vector is written both as an element of U and as an element of W, so it lies in $U \cap W$ and is a combination of the v's alone. That forces the u-coefficients to be 0, and then the rest.



$$\dim(U+W) = r + k + l = (r+k) + (r+l) - r$$

A basis of $U \cap W$ (middle) extended to a basis of U (left two groups) and of W (right two groups). All three groups together are a basis of $U+W$.

Proof. 1. Extend $v_1, \ldots, v_r$ to bases of U and of W as stated (Prop. 11 in U and in W).

2. The combined list spans $U+W$: every $u+w$ is a combination of it.

3. Independence. Suppose $\sum \lambda_i v_i + \sum \mu_j u_j + \sum \nu_t w_t = 0$. Then

$$x = \sum \lambda_i v_i + \sum \mu_j u_j = -\sum \nu_t w_t.$$

The first expression is in U, the second in W, so $x \in U \cap W$.

4. So $x = \sum \alpha_i v_i$ for some α_i. Then $\sum \alpha_i v_i + \sum \nu_t w_t = 0$ is a relation in the basis of W, so every $\nu_t = 0$.

5. The original relation is now $\sum \lambda_i v_i + \sum \mu_j u_j = 0$, a relation in the basis of U, so every $\lambda_i = 0$ and $\mu_j = 0$.

6. Count: $\dim(U \cap W) = r$, $\dim U = r+k$, $\dim W = r+l$, $\dim(U+W) = r+k+l$.

□

Proposition (Complement criterion)

For subspaces U, W of a finite-dimensional V, any two of the following imply the third (and then $V = U \oplus W$):

(a) $\dim U + \dim W = \dim V$ (b) $U \cap W = \{0\}$ (c) $U + W = V$.

Proof. All three cases use the formula $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. Note that (b) says $\dim(U \cap W) = 0$, and (c) says $\dim(U+W) = \dim V$ (a subspace of V with the same dimension as V is V, Prop. 14).

1. *(a) and (b) give (c).* The formula gives $\dim(U+W) = \dim U + \dim W - 0 = \dim V$. So $U+W = V$.

2. *(a) and (c) give (b).* The formula gives $\dim V = \dim V - \dim(U \cap W)$. So $\dim(U \cap W) = 0$, and a space of dimension 0 is $\{0\}$.

3. *(b) and (c) give (a).* The formula gives $\dim V = \dim U + \dim W - 0$.

When all three hold, $U + W = V$ and $U \cap W = \{0\}$, which is the definition of $V = U \oplus W$ (W2). □

Example (Ex. 19: two planes in $\mathbb{F}^3$)

$U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$, $W = \operatorname{span}\big((1,2,2),(2,2,1)\big)$.

1. Is $(1,2,2) \in U$? $(1,2,2) = \lambda(1,1,0) + \mu(0,1,1)$ needs $\lambda = 1$ (first coordinate) and $\mu = 2$ (third), but then the second coordinate is $3 \neq 2$. No.
2. So $(1,1,0),(0,1,1),(1,2,2)$ are independent: three independent vectors in $\mathbb{F}^3$, a basis (Prop. 13). Hence $U + W = \mathbb{F}^3$.
3. $\dim(U \cap W) = 2 + 2 - 3 = 1$: the planes meet in a line.
4. To find the line, solve $a(1,1,0)+b(0,1,1) = c(1,2,2)+d(2,2,1)$: $a = c+2d$, $a+b = 2c+2d$, $b = 2c+d$. Adding the first and third: $a+b = 3c+3d$, so $c+d = 0$. Take $c = 1$, $d = -1$: the vector is $(1,2,2)-(2,2,1) = (-1,0,1)$. So $U \cap W = \operatorname{span}\big((-1,0,1)\big)$.

- Ex. 20: for three subspaces, apply the formula to $U_1 + U_2$ and U_3, then to U_1 and U_2:

$$\dim(U_1 + U_2 + U_3) = \dim U_1 + \dim U_2 + \dim U_3 - \dim(U_1 \cap U_2) - \dim\big((U_1 + U_2) \cap U_3\big).$$

  There is no formula using only the pairwise intersections: three lines in $\mathbb{F}^2$ have all pairwise intersections $\{0\}$, and $\dim(U_1 + U_2 + U_3) = 2 \neq 3$.

Proposition (Prop. 17 / LADR 3.94)

If the sum $U_1 \oplus \cdots \oplus U_m$ is direct, then $\dim(U_1 \oplus \cdots \oplus U_m) = \dim U_1 + \cdots + \dim U_m$.

Proof. Induction on m. Write $S = U_1 + \cdots + U_{m-1}$. By the theorem,

$$\dim(S + U_m) = \dim S + \dim U_m - \dim(S \cap U_m),$$

and the last intersection is $\{0\}$ by the definition of a direct sum (W2). So $\dim(S + U_m) = \dim S + \dim U_m$. The sum $S = U_1 + \cdots + U_{m-1}$ is also direct, so by the induction hypothesis $\dim S = \dim U_1 + \cdots + \dim U_{m-1}$. Adding $\dim U_m$ gives the formula for m subspaces. □

Computing with lists in $\mathbb{F}^n$.

1. *Independent?* Put the vectors as the columns of a matrix and row reduce. Independent $\iff$ every column has a pivot. The columns with pivots always form a basis of the span.
2. *Basis of $U + W$:* put the two bases together and delete redundant vectors (Prop. 9).
3. *Basis of $U \cap W$:* set a general combination of the basis of U equal to a general combination of the basis of W and solve, as in Ex. 19. First get its dimension from the formula so you know how many vectors to expect.

Shortcuts

- **Right number of vectors**: with $\dim V$ vectors, check only independence or only spanning (Prop. 13).
- **Too many or too few**: more than $\dim V$ vectors are dependent; fewer than $\dim V$ do not span (Prop. 10). No computation.
- **Two vectors** are dependent exactly when one is a multiple of the other.
- **Polynomials of different degrees** are independent.
- **Direct sum by counting**: if $U + W = V$ and $\dim U + \dim W = \dim V$, the sum is direct (complement criterion); no need to compute $U \cap W$ (HW2 P5).
- **Equality of subspaces**: $U \subseteq W$ and $\dim U = \dim W$ give $U = W$ (Prop. 14).

- **Subspace given by k independent linear conditions** in an n-dimensional space: expect dimension $n-k$ (proved next week by rank–nullity); use this to know when you have found enough basis vectors.

Traps

- Independence of each list separately says nothing about sums: v and $-v$ are each independent lists, but $v+(-v)=0$ (HW2 P1).
- "v_j is not a multiple of any other vector in the list" does not give independence for three or more vectors: $(1,0),(0,1),(1,1)$.
- A basis of V need not contain a basis of a given subspace U: e_1, e_2 contains no basis of $\operatorname{span}((1,1))$.
- $\dim(U_1+U_2+U_3)$ has no inclusion–exclusion formula.
- The dimension depends on the field: $\dim_{\mathbb{R}} \mathbb{C} = 2$.
- Prop. 13 has no infinite-dimensional version: $\{e_1, e_2, \dots\}$ in $\mathbb{F}^\infty$ is an infinite independent set in an infinite-dimensional space, and it is not a basis.

§3 Problem set

HW2 P1 (LADR 2A.11). Prove or give a counterexample: if $v_1,\dots,v_n$ and $w_1,\dots,w_n$ are linearly independent lists in V, then $v_1+w_1,\dots,v_n+w_n$ is linearly independent.

Hint: try $n=1$.

HW2 P2. Let $\mathbb{F}^\infty$ be the space of all sequences $(a_1,a_2,\dots)$, $a_j\in\mathbb{F}$, and e_j the sequence with 1 in position j and 0 elsewhere. (a) Verify that $\mathbb{F}^\infty$ is a vector space. (b) Prove that it is infinite-dimensional. (c) Prove that $e_1, e_2, \dots$ do not span $\mathbb{F}^\infty$.

Hint: (b) Prop. 10: a finite spanning list bounds the length of independent lists. (c) a finite combination has only finitely many nonzero entries.

HW2 P3 (LADR 2B.5). Suppose V is finite-dimensional and U, W are subspaces with $V = U+W$. Prove that V has a basis all of whose vectors belong to $U\cup W$.

Hint: a spanning list contains a basis (Prop. 9).

HW2 P4. (a) Prove that every nonzero proper subspace of $\mathbb{F}^2$ has the form $\operatorname{span}(v)$ for some $v\neq 0$. (b) Let $v, w\in\mathbb{F}^2$. Prove that v, w is a basis of $\mathbb{F}^2$ if and only if neither is a scalar multiple of the other.

Hint: dimensions of subspaces (Prop. 14) and Prop. 13.

HW2 P5 (LADR 2C.12). Suppose U, W are subspaces of $\mathbb{R}^8$ with $\dim U = 3$, $\dim W = 5$ and $U+W=\mathbb{R}^8$. Prove that $\mathbb{R}^8 = U\oplus W$.

Hint: the dimension formula.

HW2 P6. Let $U=\{p\in\mathcal{P}_4(\mathbb{R}) : p(1)=p'(1)=0\}$. (a) Prove that U is a subspace. (b) Find a basis of U and determine $\dim U$ directly, without rank–nullity.

Hint: write p in the basis $1,(x-1),\dots,(x-1)^4$ of Ex. 18.

R1 (Practice Midterm 1, Question 1). Let $u_1=(1,1,0,0)$, $u_2=(0,1,1,0)$, $w_1=(0,0,1,1)$, $w_2=(1,0,0,-1)$, $U=\operatorname{span}(u_1,u_2)$, $W=\operatorname{span}(w_1,w_2)$. (a) Prove that $\mathbb{R}^4 = U\oplus W$. (b) Write $(2,1,3,0)$ as $u+w$ with $u\in U$, $w\in W$.

R2 (Midterm 1 review, Exercise 1). Let U be a subspace of a finite-dimensional V. Prove that there is a subspace W with $U\oplus W = V$.

R3 (Midterm 1 review, Exercise 3). Let $V=\{p\in\mathcal{P}_2(\mathbb{R}) : p'(1)=p(2)\}$. Construct a basis of V and prove it is a basis.